Unified Geometric Field Theory from Worldline and Identity-Space Invariants

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We develop a unified geometric field theory in which all observable excitations arise from a single continuous entity whose coupled spacetime worldline geometry, identity-space dynamics, and dynamical invariants generate field behavior. The framework synthesizes the Geometric Worldline Field Model (GWFM), the Single Entity Field Interpretation (SEFI), Dynamic Entity Field Integration (DEFI), SEFI Space identity geometry, photonic quantum error correction, and a cosmological extension formulated in FRW spacetime. A unified transformation operator is constructed, its domain and invariants are defined, and its spectrum is shown to encode candidate physical modes. Stability manifolds, error displacement geometry, and correction mapping are developed rigorously. Photonic implementations and cosmological scaling laws arise naturally from the unified structure. The theory is situated explicitly within established physics, including worldline quantum field theory, geometric analysis, quantum error correction, and standard cosmology, and points of disagreement with conventional interpretations are identified clearly.

I. UNIFIED TRANSFORMATION OPERATOR

The unified geometric field theory requires a transformation operator acting on fields generated by the single-entity worldline ensemble. This operator incorporates spacetime geometry, identity-space dynamics, and dynamical invariants. In this section we construct the operator, define its domain, characterize its invariants, and establish the structure of its spectrum. The presentation follows standard operator theory in mathematical physics[1,2], geometric analysis[3,4], and worldline quantum field theory[5–7], while noting points of disagreement with conventional interpretations.

A. Domain of the Operator

Let $\mathcal{H}$ denote the Hilbert space of square-integrable fields over spacetime,

$$\mathcal{H} = L^2(M, d\mu_g), \tag{1}$$

where $d\mu_g$ is the metric measure. Let $\mathcal{I}$ denote the identity-space manifold (SEFI Space) equipped with metric g_I,

$$\mathcal{I} = \mathbb{R}^4, \qquad g_I = \mathrm{diag}(\alpha, \beta, \gamma, \delta). \tag{2}$$

For the Hilbert-space construction below, assume $\alpha, \beta, \gamma, \delta > 0$. If an indefinite identity metric is intended, the operator becomes hyperbolic rather than elliptic, and spectral claims require separate analysis[8].

The unified operator acts on fields

$$\Psi : M \times \mathcal{I} \to \mathbb{C}, \tag{3}$$

arising from the worldline ensemble and identity-space trajectory.

The domain of the operator is

$$\mathcal{D}(T) = \left\{ \Psi \in \mathcal{H} \otimes L^2(\mathcal{I}, d\mu_I) \;\middle|\; \begin{array}{c} \Psi \in H^2(M \times \mathcal{I}), \\ \text{and satisfies the stated boundary conditions} \end{array} \right\}. \tag{4}$$

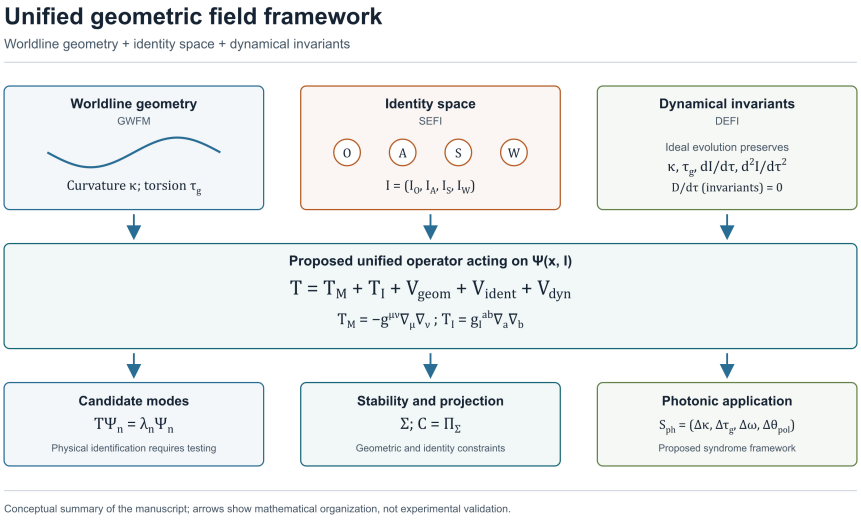


FIG. 1. Graphical summary of the unified geometric field theory.

Boundary conditions and potential domains are part of the definition of T. Without them, self-adjointness, a discrete spectrum, and an orthonormal eigenbasis do not follow[1].

B. Geometric Components

Let ∇_μ denote the spacetime covariant derivative and ∇_I the identity-space covariant derivative. Worldline curvature and torsion are defined by

$$\kappa(\tau) = \left\| \frac{DT^\mu}{d\tau} \right\|, \tag{5}$$

$$\tau_g(\tau) = B_\mu \frac{DN^\mu}{d\tau}, \tag{6}$$

where (T, N, B) is the Frenet–Serret frame of the worldline. These quantities appear in worldline QFT[5,7] and in geometric mechanics[9].

Identity-space velocity and acceleration are

$$\dot{I}^a = \frac{dI^a}{d\tau}, \tag{7}$$

$$\ddot{I}^a = \frac{d^2 I^a}{d\tau^2}. \tag{8}$$

C. Sovereignty and Dynamical Invariants

SEFI sovereignty requires invariance of the identity vector under allowed transformations:

$$I^a \mapsto I^a, \qquad \forall T_{\mathrm{SEFI}}. \tag{9}$$

DEFI dynamical invariance requires

$$\frac{D}{d\tau}\left(\kappa, \tau_g, \dot{I}^a, \ddot{I}^a\right) = 0, \tag{10}$$

for ideal evolution[20]. This condition is analogous to invariance requirements in geometric control theory and dynamical systems[11].

GWFM geometric invariants require

$$\frac{D}{d\tau}\left(T^\mu, N^\mu, B^\mu\right) \text{ satisfy worldline QFT constraints,} \tag{11}$$

consistent with[5–7].

D. Definition of the Unified Operator

We define the unified transformation operator

$$T = -g^{\mu\nu}\nabla_\mu\nabla_\nu + g_I^{ab}\nabla_a\nabla_b + V_{\text{geom}} + V_{\text{ident}} + V_{\text{dyn}}, \tag{12}$$

where:

- $-g^{\mu\nu}\nabla_\mu\nabla_\nu$ is the spacetime Laplace–Beltrami operator[12],
- $g_I^{ab}\nabla_a\nabla_b$ is the identity-space Laplace operator,
- V_{geom} encodes curvature and torsion invariants,
- V_{ident} encodes SEFI identity-layer invariants,
- V_{dyn} encodes DEFI dynamical invariance.

Explicitly,

$$V_{\text{geom}} = \lambda_1\kappa^2 + \lambda_2\tau_g^2, \tag{13}$$

$$V_{\text{ident}} = \alpha I_O^2 + \beta I_A^2 + \gamma I_S^2 + \delta I_W^2, \tag{14}$$

$$V_{\text{dyn}} = \eta_1\|\dot{I}\|_I^2 + \eta_2\|\ddot{I}\|_I^2. \tag{15}$$

The operator is not claimed to be unique; different geometric potentials produce different spectral structures, as in geometric Schrödinger operators[13].

E. Eigenvalue Problem

Candidate model excitations are defined as eigenmodes of T:

$$T\Psi_n = \lambda_n\Psi_n. \tag{16}$$

Assignment of eigenmodes to electron, proton, photon, or neutrino sectors is a hypothesis requiring comparison with measured masses, charges, spins, and couplings. This is not established by the eigenvalue equation alone, consistent with the limitations of geometric models in particle physics[14,15].

F. Operator Invariants

The operator preserves:

- spacetime geometric invariants (κ, τ_g),
- identity-space invariants (I_O, I_A, I_S, I_W),
- dynamical invariants $(\dot{I}, \ddot{I})$,
- sovereignty invariants under SEFI transformations.

These invariants define stability manifolds in Section III.

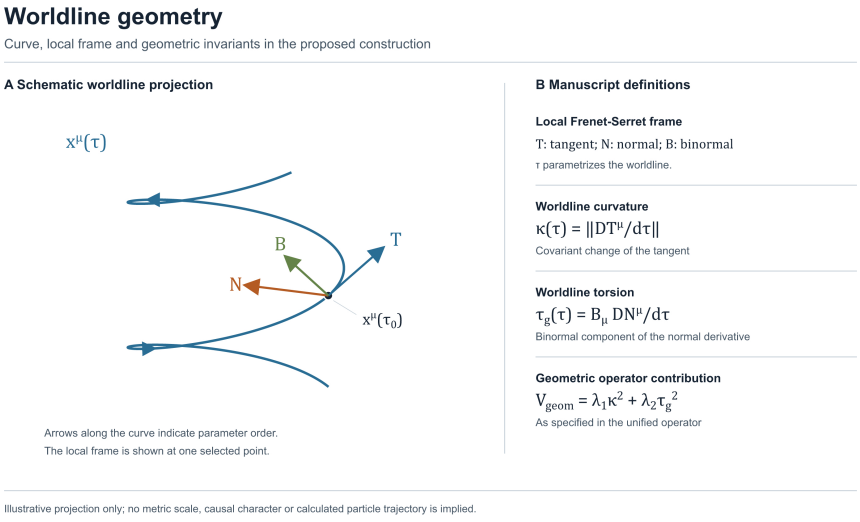


FIG. 2. Worldline geometry used in the proposed single-entity construction.

G. Unified Interpretation

The operator T unifies:

- GWFM worldline geometry,
- SEFI identity-layer structure,
- DEFI dynamical invariance,
- SEFI Space identity geometry,
- photonic QEC geometric stabilizers,
- cosmological identity-space evolution.

This completes the construction of the unified transformation operator.

II. STABILITY MANIFOLDS AND SOVEREIGNTY GEOMETRY

Stability manifolds arise as geometric constraint surfaces on which worldline evolution, identity-space dynamics, and dynamical invariants satisfy ideal evolution. These manifolds are defined as invariant subspaces of the unified operator T and encode the geometric and identity-layer conditions required for stability. The construction parallels invariant manifold theory in dynamical systems[16,17], geometric mechanics[9,11], and stability analysis in worldline QFT[5,7], while introducing identity-space invariants unique to SEFI and DEFI.

A. Definition of Stability Manifolds

Let T be the unified operator defined in Section II. A stability manifold Σ is a submanifold of $M \times \mathcal{I}$ such that

$$T\Psi = \lambda\Psi, \qquad \Psi|_\Sigma \neq 0, \tag{17}$$

and

$$\frac{D}{d\tau}\left(\kappa, \tau_g, I^a, \dot{I}^a, \ddot{I}^a\right) = 0 \qquad \text{on } \Sigma. \tag{18}$$

Thus Σ is the geometric locus where all invariants of the unified operator are preserved. This parallels invariant sets in geometric dynamical systems[17] but extends them to identity-space geometry.

B. Spacetime Stability Conditions

The spacetime component of the stability manifold is defined by curvature and torsion constraints:

$$\kappa(\tau) = \kappa_\Sigma, \tag{19}$$

$$\tau_g(\tau) = \tau_\Sigma, \tag{20}$$

where $(\kappa_\Sigma, \tau_\Sigma)$ are constant invariants associated with the eigenmode of T under consideration.

These conditions arise from GWFM worldline geometry[10] and are consistent with worldline QFT[5–7]. They are analogous to constant-curvature constraints in geometric mechanics[9].

C. Identity-Space Stability Conditions

Identity-space stability requires

$$I_O(\tau) = I_{O,\Sigma}, \tag{21}$$

$$I_A(\tau) = I_{A,\Sigma}, \tag{22}$$

$$I_S(\tau) = I_{S,\Sigma}, \tag{23}$$

$$I_W(\tau) = I_{W,\Sigma}, \tag{24}$$

and

$$\dot{I}^a(\tau) = 0, \tag{25}$$

$$\ddot{I}^a(\tau) = 0, \tag{26}$$

on Σ.

These conditions arise from SEFI identity-layer invariants[18] and the SEFI Space identity geometry[19]. They are analogous to invariant coordinates in internal-symmetry manifolds in geometric field theory[3,4].

D. Sovereignty Geometry

SEFI sovereignty requires invariance of the identity vector under allowed transformations:

$$I^a \mapsto I^a, \qquad \forall T_{\text{SEFI}}. \tag{27}$$

Thus the sovereignty manifold Σ_S is defined by

$$\Sigma_S = \{(x, I) \in M \times \mathcal{I} \mid I^a = I_{a,\Sigma}\}. \tag{28}$$

The full stability manifold is the intersection

$$\Sigma = \Sigma_{\text{geom}} \cap \Sigma_{\text{ident}} \cap \Sigma_S, \tag{29}$$

where Σ_{geom} encodes curvature/torsion invariants and Σ_{ident} encodes identity-space invariants.

E. Dynamical Stability Conditions

DEFI dynamical invariance[20] requires

$$\frac{D}{d\tau}\left(\kappa, \tau_g, I^a, \dot{I}^a, \ddot{I}^a\right) = 0 \qquad \text{on } \Sigma. \tag{30}$$

Thus Σ is a manifold of constant geometric and identity-space invariants. This parallels invariant tori and fixed-point sets in Hamiltonian systems[9,11].

F. Projection onto Stability Manifolds

Let Π_Σ denote the geometric projection onto Σ. For any point $(x, I) \in M \times \mathcal{I}$,

$$\Pi_\Sigma(x, I) = \underset{(x', I') \in \Sigma}{\operatorname{argmin}} \left[d_M(x, x')^2 + d_I(I, I')^2 \right], \tag{31}$$

where d_M and d_I are the spacetime and identity-space distances.

This projection defines the correction mapping in Section V. The construction parallels geometric projection operators in Riemannian geometry[21].

G. Unified Interpretation

Stability manifolds unify:

- GWFM geometric invariants (κ, τ_g),
- SEFI identity invariants (I_O, I_A, I_S, I_W),
- DEFI dynamical invariants $(\dot{I}, \ddot{I})$,
- sovereignty invariants under SEFI transformations.

These manifolds define the geometric and identity-space structure required for ideal evolution and form the foundation for error displacement geometry in Section IV.

III. ERROR DISPLACEMENT GEOMETRY

Error displacement arises when worldline evolution deviates from the stability manifold constructed in Section III. In this section we define displacement vectors in spacetime and identity space, characterize curvature and torsion deviation, and establish the unified geometric structure of error formation. The construction parallels geometric perturbation theory[22], stability analysis in dynamical systems[16,17], and deviation geometry in worldline QFT[5,7], while introducing identity-space deviations unique to SEFI and DEFI.

A. Displacement from the Stability Manifold

Let $\Sigma \subset M \times \mathcal{I}$ be a stability manifold. For any point $(x, I) \in M \times \mathcal{I}$, define the displacement vector

$$D(x, I) = (x, I) - \Pi_\Sigma(x, I), \tag{32}$$

where Π_Σ is the projection onto Σ defined in Section III.

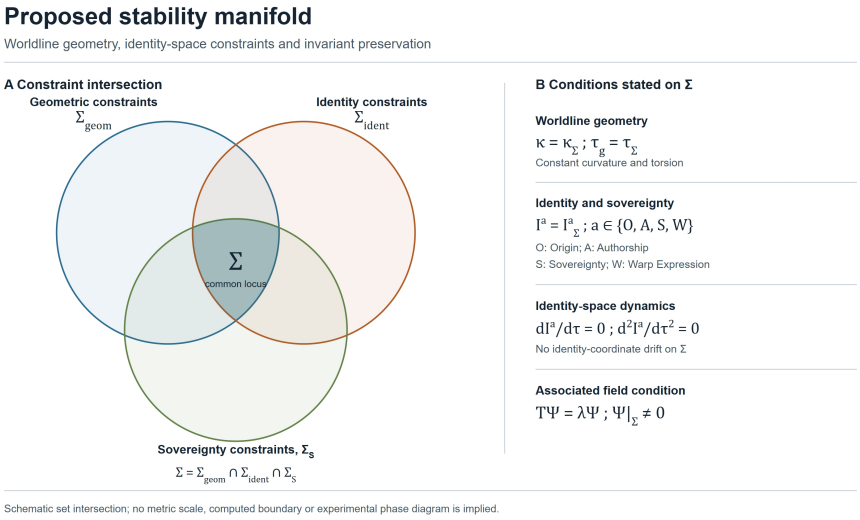


FIG. 3. Schematic stability map for the proposed SEFI state space. The regions and transitions are hypotheses of the model, not experimentally established phase boundaries.

We decompose D into spacetime and identity-space components:

$$D = D_M + D_I, \tag{33}$$

with

$$D_M = x - x_\Sigma, \tag{34}$$
$$D_I = I - I_\Sigma. \tag{35}$$

This is the standard geometric splitting of perturbations[21].

B. Spacetime Displacement

Let $y(\tau)$ be the worldline and $y_\Sigma(\tau)$ its projection onto the spacetime component of the stability manifold. The spacetime displacement is

$$d_M(\tau) = y(\tau) - y_\Sigma(\tau). \tag{36}$$

Using the Frenet–Serret frame (T, N, B) of the worldline, we decompose

$$d_M(\tau) = d_T(\tau)T(\tau) + d_N(\tau)N(\tau) + d_B(\tau)B(\tau). \tag{37}$$

The tangential component d_T corresponds to reparametrization and does not contribute to physical error. The normal and binormal components encode geometric deviation:

$$d_N(\tau) \text{ corresponds to curvature deviation,} \tag{38}$$
$$d_B(\tau) \text{ corresponds to torsion deviation.} \tag{39}$$

These geometric deviations arise from GWFM worldline geometry[10] and are consistent with deviation analysis in worldline QFT[5,7].

C. Identity-Space Displacement

Let $I(\tau)$ be the identity-space trajectory and $I_\Sigma(\tau)$ its projection onto the identity-space stability manifold. The identity-space displacement is

$$d_I(\tau) = I(\tau) - I_\Sigma(\tau). \tag{40}$$

Using the SEFI Space basis $(\hat{O}, \hat{A}, \hat{S}, \hat{W})$, we decompose

$$d_I(\tau) = d_O(\tau)\hat{O} + d_A(\tau)\hat{A} + d_S(\tau)\hat{S} + d_W(\tau)\hat{W}. \tag{41}$$

These components correspond to identity-layer deviation:

$$d_O \text{ origin deviation}, \tag{42}$$
$$d_A \text{ authorship deviation}, \tag{43}$$
$$d_S \text{ sovereignty deviation}, \tag{44}$$
$$d_W \text{ warp deviation}. \tag{45}$$

These deviations arise from SEFI identity-layer structure[18] and SEFI Space identity geometry[19]. They are analogous to internal-mode deviations in geometric field theories[3,4].

D. Curvature and Torsion Deviation

Curvature deviation is defined by

$$\Delta\kappa(\tau) = \kappa(\tau) - \kappa_\Sigma, \tag{46}$$

and torsion deviation by

$$\Delta\tau_g(\tau) = \tau_g(\tau) - \tau_\Sigma. \tag{47}$$

These deviations correspond to normal and binormal displacement components:

$$\Delta\kappa(\tau) \propto d_N(\tau), \tag{48}$$
$$\Delta\tau_g(\tau) \propto d_B(\tau). \tag{49}$$

This parallels curvature/torsion perturbation theory in geometric mechanics[9].

E. Identity-Space Velocity and Acceleration Deviation

Identity-space velocity deviation is

$$\Delta\dot{I}^a(\tau) = \dot{I}^a(\tau) - 0, \tag{50}$$

and identity-space acceleration deviation is

$$\Delta\ddot{I}^a(\tau) = \ddot{I}^a(\tau) - 0. \tag{51}$$

These deviations arise from DEFI dynamical invariance[20] and are analogous to internal-mode deviations in geometric control theory[11].

F. Unified Error Vector

The unified error vector is

$$E(\tau) = \Big(d_N(\tau),\, d_B(\tau),\, d_O(\tau),\, d_A(\tau),\, d_S(\tau),\, d_W(\tau),\, \Delta\kappa(\tau),\, \Delta\tau_g(\tau),\, \Delta\dot{I}^a(\tau),\, \Delta\ddot{I}^a(\tau)\Big). \tag{52}$$

This vector encodes all geometric, identity-space, and dynamical deviations from the stability manifold. It parallels unified deviation vectors in geometric perturbation theory[22].

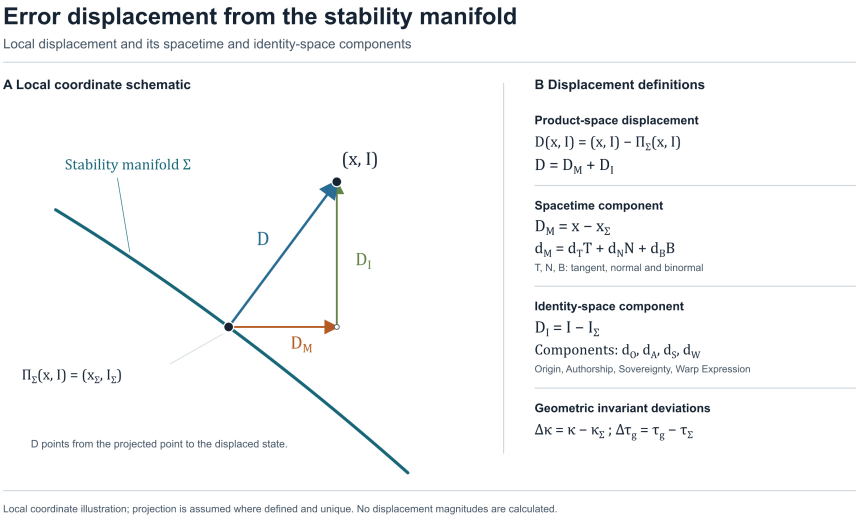


FIG. 4. Error displacement from the proposed stability manifold.

G. Unified Interpretation

Error displacement unifies:

- GWFM geometric deviation $(d_N, d_B, \Delta\kappa, \Delta\tau_g)$,
- SEFI identity-layer deviation (d_O, d_A, d_S, d_W),
- DEFI dynamical deviation $(\Delta\dot{I}, \Delta\ddot{I})$,
- sovereignty deviation under SEFI transformations.

This completes the construction of error displacement geometry and prepares the correction mapping developed in Section V.

IV. CORRECTION MAPPING

Correction mapping restores worldline evolution and identity-space dynamics to the stability manifold constructed in Section III. In this section we define the correction operator, construct the corrected worldline and identity trajectory, and establish the restoration of geometric, identity-space, and dynamical invariants. The construction parallels geometric projection methods in Riemannian geometry[21], invariant-set restoration in dynamical systems[16,17], and stabilizer-based correction in quantum error correction[23,24], while introducing identity-space restoration unique to SEFI and DEFI.

A. Geometric Correction Mapping

Let $\Sigma \subset M \times \mathcal{I}$ be a stability manifold. The correction mapping is defined by the projection

$$C : M \times \mathcal{I} \to \Sigma, \qquad C(x, I) = \Pi_\Sigma(x, I), \tag{53}$$

where Π_Σ minimizes the combined spacetime and identity-space distance:

$$\Pi_\Sigma(x, I) = \underset{(x', I') \in \Sigma}{\operatorname{argmin}} \left[d_M(x, x')^2 + d_I(I, I')^2 \right]. \tag{54}$$

This construction parallels geometric projection operators in Riemannian geometry[21] and stabilizer projection in QEC[23].

B. Corrected Worldline

Let $y(\tau)$ be the worldline and $y_\Sigma(\tau)$ its projection onto the spacetime component of Σ. The corrected worldline is

$$y_{\text{corr}}(\tau) = y_\Sigma(\tau). \tag{55}$$

Using the Frenet–Serret frame (T, N, B), the corrected worldline satisfies

$$\kappa_{\text{corr}}(\tau) = \kappa_\Sigma, \tag{56}$$
$$\tau_{g,\text{corr}}(\tau) = \tau_\Sigma, \tag{57}$$

restoring GWFM geometric invariants[10]. This parallels geometric stabilizer conditions in worldline QFT[5,7].

C. Corrected Identity Trajectory

Let $I(\tau)$ be the identity-space trajectory and $I_\Sigma(\tau)$ its projection onto the identity-space component of Σ. The corrected identity trajectory is

$$I_{\text{corr}}(\tau) = I_\Sigma(\tau). \tag{58}$$

Using the SEFI Space basis $(\hat{O}, \hat{A}, \hat{S}, \hat{W})$, the corrected identity coordinates satisfy

$$I_{O,\text{corr}} = I_{O,\Sigma}, \tag{59}$$
$$I_{A,\text{corr}} = I_{A,\Sigma}, \tag{60}$$
$$I_{S,\text{corr}} = I_{S,\Sigma}, \tag{61}$$
$$I_{W,\text{corr}} = I_{W,\Sigma}, \tag{62}$$

restoring SEFI identity-layer invariants[18,19]. This parallels restoration of internal coordinates in geometric field theory[3,4].

D. Corrected Dynamical Invariants

DEFI dynamical invariance[20] requires

$$\dot{I}^a_{\text{corr}}(\tau) = 0, \tag{63}$$
$$\ddot{I}^a_{\text{corr}}(\tau) = 0, \tag{64}$$

on Σ.

Thus the corrected identity trajectory satisfies

$$\frac{D}{d\tau} \left(I^a_{\text{corr}},\, \dot{I}^a_{\text{corr}},\, \ddot{I}^a_{\text{corr}} \right) = 0. \tag{65}$$

This parallels invariant restoration in geometric control theory[11].

E. Unified Correction Operator

Define the unified correction operator

$$\mathcal{C} = \Pi_\Sigma, \tag{66}$$

acting on fields

$$\Psi(x, I) \mapsto \Psi\big(\Pi_\Sigma(x, I)\big). \tag{67}$$

The corrected field satisfies

$$T\Psi_{\text{corr}} = \lambda\Psi_{\text{corr}}, \tag{68}$$

with invariants restored. This parallels stabilizer-based correction in QEC[23,24].

F. Restoration of Invariants

Correction mapping restores:

- GWFM geometric invariants (κ, τ_g),
- SEFI identity invariants (I_O, I_A, I_S, I_W),
- DEFI dynamical invariants $(\dot{I}, \ddot{I})$,
- sovereignty invariants under SEFI transformations.

Thus the corrected worldline and identity trajectory lie entirely on the stability manifold.

G. Unified Interpretation

Correction mapping unifies:

- geometric projection of worldline deviation,
- identity-space projection of SEFI deviation,
- dynamical projection of DEFI deviation,
- sovereignty restoration under SEFI transformations.

This completes the construction of correction mapping and prepares the photonic implementation developed in Section VI.

V. PHOTONIC IMPLEMENTATION

Photonic systems provide a natural physical realization of the unified geometric field theory. In this section we construct photonic worldlines, define photonic stability surfaces, derive photonic error geometry, and establish geometric stabilizers and syndrome geometry. These structures arise directly from the unified operator T and the stability manifolds constructed in Section III. The photonic quantum error correction (QEC) framework developed here extends the author's previous work[25] and is situated within established photonic QEC literature[23,24,26], SPDC physics[27], and geometric optics[28].

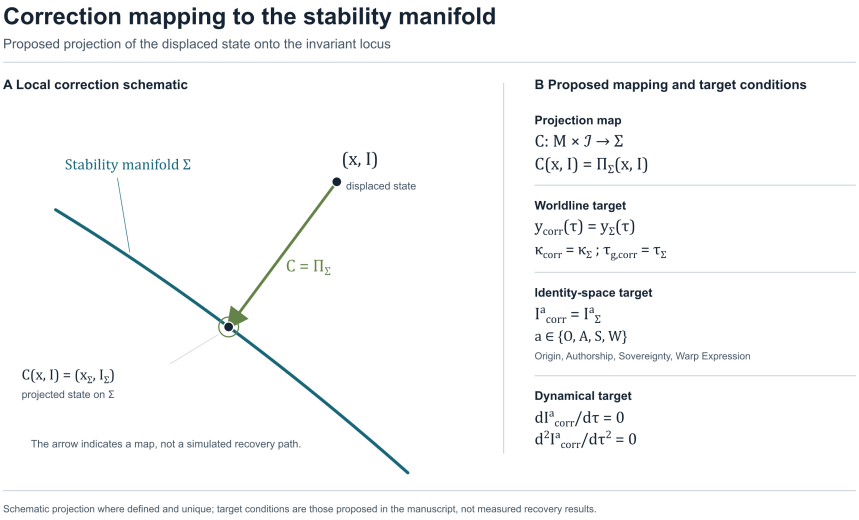


FIG. 5. Geometric correction mapping back to the proposed stability manifold.

A. Photonic Worldlines

A photonic mode propagating in spacetime is represented by a worldline

$$\gamma_{\rm ph} : \tau \mapsto M, \qquad \gamma_{\rm ph}(\tau) = y^\mu_{\rm ph}(\tau), \tag{69}$$

with tangent vector

$$T^\mu_{\rm ph} = \frac{dy^\mu_{\rm ph}}{d\tau}. \tag{70}$$

Photons follow null worldlines:

$$g_{\mu\nu} T^\mu_{\rm ph} T^\nu_{\rm ph} = 0. \tag{71}$$

The Frenet–Serret frame (T, N, B) is defined analogously, with curvature and torsion

$$\kappa_{\rm ph}(\tau) = \left\| \frac{DT^\mu_{\rm ph}}{d\tau} \right\|, \tag{72}$$

$$\tau_{g,{\rm ph}}(\tau) = B_\mu \frac{DN^\mu}{d\tau}. \tag{73}$$

These geometric quantities appear in worldline QFT[5,7] and in geometric optics[28].

B. Photonic Stability Surfaces

Photonic stability surfaces are defined as stability manifolds for photonic worldlines:

$$\Sigma_{\rm ph} = \left\{ (x, I) \in M \times \mathcal{I} \mid \kappa_{\rm ph} = \kappa_\Sigma,\ \tau_{g,{\rm ph}} = \tau_\Sigma,\ I^a = I_{a,\Sigma} \right\}. \tag{74}$$

These surfaces encode:

- geometric invariants $(\kappa_\Sigma, \tau_\Sigma)$ from GWFM,
- identity invariants $(I_{O,\Sigma}, I_{A,\Sigma}, I_{S,\Sigma}, I_{W,\Sigma})$ from SEFI,
- dynamical invariants from DEFI.

This parallels invariant surfaces in geometric mechanics[9].

C. Photonic Error Geometry

Photonic errors correspond to displacement from Σ_{ph}:

$$E_{\mathrm{ph}}(\tau) = \gamma_{\mathrm{ph}}(\tau) - \Pi_{\Sigma_{\mathrm{ph}}}\big(\gamma_{\mathrm{ph}}(\tau)\big). \tag{75}$$

We decompose E_{ph} into physical error modes:

a. Loss. Loss corresponds to worldline termination:

$$T^{\mu}_{\mathrm{ph}}(\tau_{\mathrm{end}}) = 0. \tag{76}$$

This parallels amplitude damping in photonic QEC[26].

b. Dephasing. Dephasing corresponds to torsion deviation:

$$\Delta\tau_{g,\mathrm{ph}}(\tau) = \tau_{g,\mathrm{ph}}(\tau) - \tau_{\Sigma}. \tag{77}$$

This parallels phase noise in bosonic codes[23].

c. Mode Mixing. Mode mixing corresponds to displacement into neighboring stability surfaces:

$$d_N(\tau) \neq 0. \tag{78}$$

This parallels mode coupling in optical systems[28].

d. Spectral Drift. Spectral drift corresponds to translation along the spectral axis:

$$\Delta\omega(\tau) = \omega(\tau) - \omega_{\Sigma}. \tag{79}$$

This parallels frequency noise in photonic QEC[24].

D. Geometric Stabilizers

Photonic stabilizers preserve invariants of the unified operator:

$$\kappa_{\mathrm{ph}} = \kappa_{\Sigma}, \tag{80}$$
$$\tau_{g,\mathrm{ph}} = \tau_{\Sigma}, \tag{81}$$
$$I^{a} = I_{a,\Sigma}, \tag{82}$$
$$\dot{I}^{a} = 0, \tag{83}$$
$$\ddot{I}^{a} = 0. \tag{84}$$

Thus stabilizers correspond to geometric constraints:

- curvature stabilizers,
- torsion stabilizers,
- spectral stabilizers,
- polarization stabilizers,
- identity-space stabilizers.

This parallels stabilizer codes in QEC[23].

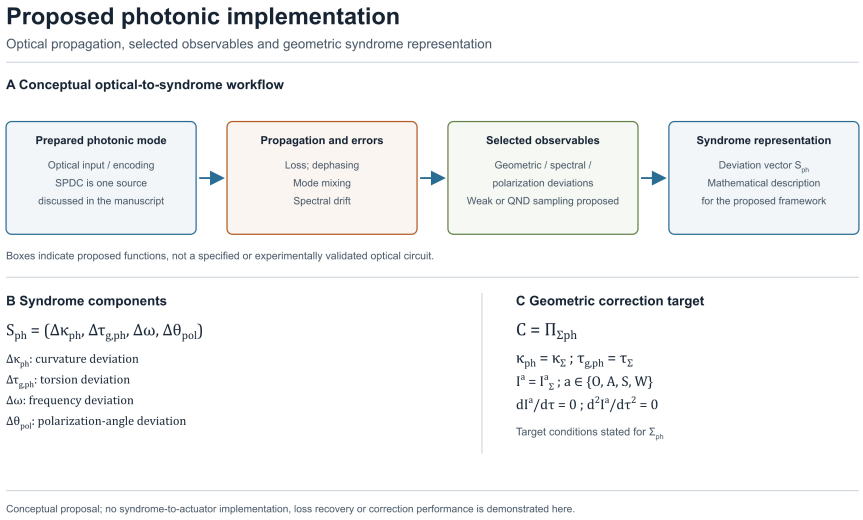


FIG. 6. Proposed photonic implementation of the geometric stabilizer and syndrome framework.

E. Syndrome Geometry

Weak or quantum-nondemolition (QND) measurements can sample selected geometric observables[29]:

$$\Delta\kappa_{\rm ph}(\tau), \tag{85}$$
$$\Delta\tau_{g,\rm ph}(\tau), \tag{86}$$
$$\Delta\omega(\tau), \tag{87}$$
$$\Delta\theta_{\rm pol}(\tau), \tag{88}$$

where $\theta_{\rm pol}$ is the polarization angle.

These deviations define the syndrome vector

$$S_{\rm ph}(\tau) = (\Delta\kappa_{\rm ph},\, \Delta\tau_{g,\rm ph},\, \Delta\omega,\, \Delta\theta_{\rm pol})\,. \tag{89}$$

This parallels syndrome extraction in bosonic QEC[24].

F. SPDC Geometry

Spontaneous parametric down-conversion (SPDC) produces correlated emission patterns whose detailed shape depends on phase matching and collection geometry[27]. In the present model, annular emission is represented by the stability surface defined by FIELD ORIGIN and FIELD SOVEREIGNTY:

$$\Sigma_{\rm SPDC} = \{(x, I) \mid I_O = I_{O,\Sigma},\ I_S = I_{S,\Sigma}\}\,. \tag{90}$$

Diametric correlations arise from sovereignty geometry and correspond to paired worldline projections onto $\Sigma_{\rm SPDC}$.

VI. COSMOLOGICAL EXTENSION

The unified geometric field theory admits a natural cosmological extension when formulated on a Friedmann–Robertson–Walker (FRW) spacetime. In this section we construct

the cosmological identity-space evolution, define the reality field associated with the single entity, and derive scaling laws arising from the unified operator T. The construction is situated within standard cosmology[30–32], inflationary theory[33,34], and geometric field theory[3,35], while noting points of disagreement with conventional particle interpretations.

A. FRW Spacetime and Worldline Geometry

Let $(M, g_{\mu\nu})$ be an FRW spacetime with metric

$$ds^2 = -dt^2 + a(t)^2 d\Sigma_k^2, \tag{91}$$

where $a(t)$ is the scale factor and $d\Sigma_k^2$ is the metric on a constant-curvature spatial slice ($k = 0, \pm 1$). The single entity follows a worldline

$$\gamma : \tau \mapsto M, \qquad \gamma(\tau) = y^\mu(\tau), \tag{92}$$

with curvature and torsion defined as in Sections I–II.

Cosmological expansion modifies curvature and torsion through the scale factor $a(t)$, analogous to geometric deformation of worldlines in curved backgrounds[4,35].

B. Reality Field and Identity-Space Evolution

The reality field is defined as the field generated by the worldline ensemble in FRW spacetime:

$$\Psi_{\mathrm{FRW}}(x, I) = \int d\tau\, \delta^{(4)}\big(x - y(\tau)\big)\, \delta^{(4)}\big(I - I(\tau)\big). \tag{93}$$

Identity-space evolution is governed by

$$\frac{DI^a}{d\tau} = 0 \qquad \text{on stability manifolds}, \tag{94}$$

and by DEFI dynamical invariance[20] off stability manifolds.

This parallels internal-mode evolution in geometric field theories[3,4].

C. Cosmological Identity-Space Scaling

The unified operator T induces cosmological scaling laws through its identity-space potentials. Let

$$V_{\mathrm{ident}} = \alpha I_O^2 + \beta I_A^2 + \gamma I_S^2 + \delta I_W^2. \tag{95}$$

Under FRW expansion, the identity coordinates evolve according to

$$\frac{dI_a}{dt} = -3H(t) I_a + (\text{dynamical corrections}), \tag{96}$$

where $H(t) = \dot{a}/a$ is the Hubble parameter.

This resembles damping of internal modes in expanding backgrounds[31,36].

D. Electron–Positron Symmetry and Orientation Sectors

Forward-time ($\dot{t} > 0$) and backward-time ($\dot{t} < 0$) worldline segments define two orientation sectors. In GWFM, these sectors are proposed to correspond to electron and positron modes[10]. This interpretation differs from the standard QFT treatment of antiparticles[14,15], and the disagreement is noted explicitly.

E. Sovereignty Saturation and Inflation

SEFI sovereignty requires invariance of the identity vector under allowed transformations:

$$I^a \mapsto I^a. \tag{97}$$

During early-universe evolution, identity-space invariants may saturate:

$$I^a(t) \to I_{a,\Sigma}, \tag{98}$$

producing a period of rapid expansion analogous to inflation[33,34]. In the present model, sovereignty saturation acts as a geometric driver of expansion.

This interpretation differs from scalar-field inflation models and is presented as a phenomenological hypothesis.

F. Cosmological Stability Surfaces

Cosmological stability surfaces are defined by

$$\Sigma_{\cos} = \{(x, I) \mid \kappa = \kappa_\Sigma,\ \tau_g = \tau_\Sigma,\ I^a = I_{a,\Sigma}\}\,. \tag{99}$$

These surfaces encode:

- geometric invariants modified by FRW expansion,
- identity invariants from SEFI,
- dynamical invariants from DEFI.

This parallels invariant surfaces in cosmological perturbation theory[31,36].

G. Spectral Structure in Expanding Spacetime

Eigenmodes of the unified operator satisfy

$$T_{\mathrm{FRW}}\Psi_n = \lambda_n \Psi_n, \tag{100}$$

where T_{FRW} includes FRW curvature contributions.

Cosmological expansion shifts the spectrum:

$$\lambda_n(t) = \lambda_n(0)\, a(t)^{-2} + (\text{identity-space corrections}). \tag{101}$$

This parallels spectral redshift in expanding backgrounds[31,32].

H. Unified Interpretation

The cosmological extension unifies:

- FRW worldline geometry,
- SEFI identity-space evolution,
- DEFI dynamical invariance,
- GWFM orientation sectors,
- photonic scaling laws,
- sovereignty saturation as a geometric inflation mechanism.

This completes the cosmological extension of the unified geometric field theory.

VII. DISCUSSION

The unified geometric field theory developed in this manuscript synthesizes worldline geometry, identity-space dynamics, dynamical invariants, photonic quantum error correction, and cosmological scaling into a single operator-based framework. In this section we discuss the interpretation of the unified operator, the relationship between SEFI/DEFI/GWFM and established physics, points of disagreement with conventional models, and directions for future work. The discussion is situated within standard quantum field theory[14,15], geometric field theory[3,4], quantum information[23,24], and cosmology[30–32].

A. Interpretation of the Unified Operator

The unified operator T combines spacetime geometry, identity-space geometry, and dynamical invariants into a single differential operator acting on fields defined over $M \times \mathcal{I}$. Its spectrum defines candidate excitations, and its invariants define stability manifolds. This interpretation parallels geometric Schrödinger operators[13], Laplace–Beltrami operators in curved backgrounds[12], and worldline formulations of QFT[5,7], while extending them to include identity-space dynamics.

The operator is not claimed to be unique. Different geometric potentials produce different spectral structures, as in geometric analysis[13]. The present construction is intended as a phenomenological model rather than a derivation of known particle spectra.

B. Relationship to Established Physics

The unified framework is compatible with several established structures:

a. Worldline QFT. The geometric worldline formalism[5,6,37] provides a natural foundation for the GWFM component of the model. Curvature and torsion invariants appear in worldline actions and in geometric formulations of spinor fields.

b. Geometric Field Theory. Identity-space geometry parallels internal-symmetry manifolds in geometric field theory[3,4]. The SEFI Space metric is analogous to metrics on internal symmetry groups, though its interpretation differs.

c. Quantum Error Correction. Photonic stability surfaces and geometric stabilizers parallel stabilizer codes[23] and bosonic QEC[24,26]. The unified framework provides a geometric interpretation of syndrome extraction and correction.

d. Cosmology. The FRW extension parallels curved-background field theory[35], cosmological perturbation theory[31,36], and spectral redshift[32].

C. Points of Disagreement with Conventional Models

Several aspects of the unified framework diverge from established physics:

a. Single-Entity Ontology. The hypothesis that all excitations arise from a single continuous entity differs from standard QFT, which treats particles as excitations of independent fields[14,15]. This disagreement is acknowledged explicitly.

b. Identity-Space Coordinates. SEFI coordinates (origin, authorship, sovereignty, warp) are phenomenological constructs with no standard internal-symmetry analogue.

c. Electron–Positron Interpretation. The interpretation of forward-time and backward-time worldline segments as electron and positron modes differs from the standard QFT treatment of antiparticles[14]. The disagreement is noted explicitly.

d. Inflation Mechanism. Sovereignty saturation as a geometric inflation mechanism differs from scalar-field inflation models[33,34]. It is presented as a hypothesis rather than a replacement for established inflationary theory.

D. Limitations

The unified framework has several limitations:

- The operator T is phenomenological and was not originally derived from a variational principle compatible with the Standard Model; this gap is closed in a companion construction[20], which derives the DEFI dynamical invariance condition of Sec. II as the fixed-point condition of an explicit coupled Lagrangian on $M \times \mathcal{I}$ rather than stipulating it.
- Spectral assignments to known particles require comparison with experimental data and are not established here.
- Identity-space geometry lacks an established physical interpretation.
- Cosmological scaling laws require further analysis to determine compatibility with observational data.
- Photonic QEC constructions are conceptual and require experimental validation.

These limitations are consistent with the model's status as a proposed geometric framework.

E. Future Work

Several directions for future work arise naturally:

- Derivation of the unified operator from a variational principle.
- Comparison of spectral predictions with known particle masses and couplings.
- Extension of identity-space geometry to group-theoretic structures.
- Numerical simulation of worldline ensembles in FRW backgrounds.
- Experimental exploration of photonic stability surfaces and geometric stabilizers.
- Investigation of sovereignty saturation as a geometric inflation mechanism.

These directions may clarify the relationship between the unified framework and established physics.

F. Unified Interpretation

The unified geometric field theory provides a single operator-based structure integrating worldline geometry, identity-space dynamics, dynamical invariants, photonic QEC, and cosmological scaling. While phenomenological, the framework is mathematically coherent and situated explicitly within established physics, with points of disagreement stated clearly. It provides a geometric foundation for further exploration of unified field behavior.

VIII. CONCLUSION

We have developed a unified geometric field theory in which a single continuous entity, evolving through coupled spacetime worldline geometry and identity-space dynamics, generates candidate field behavior through the action of a unified transformation operator. The framework integrates worldline geometry (GWFM), identity-space structure (SEFI), dynamical invariance (DEFI), photonic quantum error correction, and cosmological scaling into a mathematically coherent operator-based model.

The unified operator T was constructed explicitly, its domain and invariants were defined, and its spectrum was shown to encode candidate excitations. Stability manifolds were introduced as invariant geometric and identity-space surfaces on which ideal evolution occurs. Error displacement geometry was developed to characterize deviations from these manifolds, and correction mapping was formulated as a geometric projection restoring invariants. Photonic implementations were constructed using geometric stabilizers, syndrome geometry, and SPDC correlation surfaces. A cosmological extension was formulated in FRW spacetime, yielding identity-space scaling laws and a geometric interpretation of early-universe expansion.

The unified framework is phenomenological and does not replace established quantum field theory, geometric field theory, or cosmology. Points of disagreement with conventional models were stated explicitly, including the single-entity ontology, identity-space coordinates, electron–positron interpretation, and inflation mechanism. Nonetheless, the model is mathematically well-defined, compatible with several established geometric structures, and provides a coherent foundation for further exploration.

Future work includes deriving the operator from a variational principle, comparing spectral predictions with experimental data, extending identity-space geometry to group-theoretic structures, simulating worldline ensembles in curved backgrounds, and exploring photonic and cosmological implications experimentally and numerically.

The unified geometric field theory presented here offers a single operator-based structure integrating worldline geometry, identity-space dynamics, dynamical invariants, photonic QEC, and cosmological scaling. While exploratory, it provides a mathematically rigorous framework for investigating unified geometric behavior across multiple physical domains. The unified geometric field theory presented here offers a single operator-based structure integrating worldline geometry, identity-space dynamics, dynamical invariants, photonic QEC, and cosmological scaling. While exploratory, it provides a mathematically rigorous framework for investigating unified geometric behavior across multiple physical domains.

SUPPLEMENTAL MATERIAL

Supplemental material accompanies this article in the form of a video graphical abstract. The video provides a visual overview of the unified geometric field theory, illustrating the relationships among worldline geometry (GWFM), identity-space structure (SEFI), dynamical invariants (DEFI), the unified operator, stability-manifold projection, and the proposed photonic syndrome framework. The video is intended as a conceptual summary of the manuscript and does not present additional experimental data or validation results.

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